

ATPL EASA STUDY GUIDE | 010 - AIR LAW

Chapter 6: Instrument Departures (SID & NADP)

Syllabus Source: CAE Oxford (pp. 157-172) | Pass Target: >= 90% AviationExam

1. PANS-OPS DEPARTURE DESIGN GRADIENTS

Instrument departure procedures (ICAO Doc 8168 PANS-OPS) provide obstacle clearance from the end of the runway until the aircraft reaches the minimum en-route altitude (MEA). Design assumes all engines operating.

[*] OFFICIAL DEFINITION - Procedure Design Gradient (PDG) = 3.3%

The standard PANS-OPS Procedure Design Gradient (PDG) is 3.3% (~200 ft/NM). It consists of: 1. Obstacle Identification Surface (OIS): 2.5% 2. Obstacle Clearance Margin: 0.8% -> Total Standard PDG = 3.3%. A steeper climb gradient is published ONLY if required by obstacles or airspace.

2. STRAIGHT VS TURNING DEPARTURES

- Screen Height at DER: The departure procedure assumes the aircraft crosses the Departure End of Runway (DER) at a minimum screen height of 5 m (16 ft) with wings level.
- Straight Departure: Track does not diverge by more than 15° from runway centerline heading.
- Turning Departure: Specified whenever track requires a turn of MORE THAN 15°. No turn shall be initiated below 120 m (394 ft) above aerodrome elevation (or DER).
- Turn Design Parameters: Average bank angle: 15° (max 21°). Speed limits are specified per category.

3. NOISE ABATEMENT DEPARTURE PROCEDURES (NADP 1 VS NADP 2)

ICAO Annex 6 and PANS-OPS define two standardized noise abatement departure procedures for jet aeroplanes:

Procedure	Climb Profile & Flap Retraction Schedule
NADP 1 (Noise Close to Aerodrome)	Climb at V2 + 10 to 20 kt with take-off flaps to 800 ft (or higher). At 800 ft to 3,000 ft, reduce to climb thrust, maintain climb speed with flaps intact. At 3,000 ft, accelerate and retract flaps.
NADP 2 (Noise Distant from Aerodrome)	Climb at V2 + 10 to 20 kt to 800 ft. At 800 ft, accelerate while retracting flaps on schedule, then reduce to climb thrust. Continue climb to 3,000 ft.

4. OMNIDIRECTIONAL DEPARTURES & OBSTACLE CLEARANCES

Used when no specific SID route is published. The departure is divided into two distinct phases: - Phase 1: Aircraft climbs straight ahead on runway centerline to at least 120 m (394 ft) above aerodrome elevation. - Phase 2: Aircraft initiates turn towards the en-route track while maintaining at least 3.3% PDG. - Obstacle clearance: 90 m (295 ft) MOC provided throughout the turn area, increasing with distance from aerodrome.

5. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: What is the standard Procedure Design Gradient (PDG) under PANS-OPS for instrument departures?

[A] 2.5% [B] 3.3% [C] 5.0% [D] 6.5%

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. The standard PDG is 3.3% (consisting of 2.5% Obstacle Identification Surface + 0.8% safety margin).

Q2: What is the assumed screen height over the Departure End of Runway (DER)?

[A] 0 m [B] 5 m (16 ft) [C] 10.7 m (35 ft) [D] 15 m (50 ft)

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. PANS-OPS assumes 5 m (16 ft) screen height over DER (Note: CS-25 engine-out performance assumes 35 ft, but PANS-OPS obstacle assessment starts at 5 m).

Q3: What is the minimum altitude above aerodrome elevation before a turn may be initiated on a turning departure?

[A] 90 m (295 ft) [B] 120 m (394 ft) [C] 150 m (492 ft) [D] 300 m (984 ft)

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. No turn shall be initiated below 120 m (394 ft) above the elevation of the aerodrome.

Q4: Which noise abatement departure procedure provides noise relief for areas close to the aerodrome?

[A] NADP 1 [B] NADP 2 [C] Continuous Descent Departure [D] Balanced Field Take-off

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [A]. NADP 1 provides noise relief close to the airport by maintaining flaps and climbing steeply to 3,000 ft before accelerating.

Q5: What average bank angle is assumed in the design of turning departures under PANS-OPS?

[A] 10° [B] 15° (maximum 21°) [C] 25° [D] 30°

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. PANS-OPS departure design assumes an average bank angle of 15°, not to exceed 21°.

Q6: In an omnidirectional departure, what is the minimum height above aerodrome elevation before initiating a turn?

[A] 90 m [B] 120 m (394 ft) [C] 150 m [D] 300 m

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. In Phase 1 of an omnidirectional departure, the aircraft must climb straight ahead on runway heading to at least 120 m (394 ft) before beginning any turn.

Q7: Under what circumstances is a departure climb gradient steeper than the standard 3.3% published on a SID?

[A] Always for jet aircraft operations [B] When dictated by obstacle clearance requirements or airspace restrictions [C] Only during summer high-temperature operations [D] Whenever the runway length is less than 2,000 m

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. A gradient steeper than the standard 3.3% PDG is published only when required by obstacles in the climb path or by ATC airspace constraints.